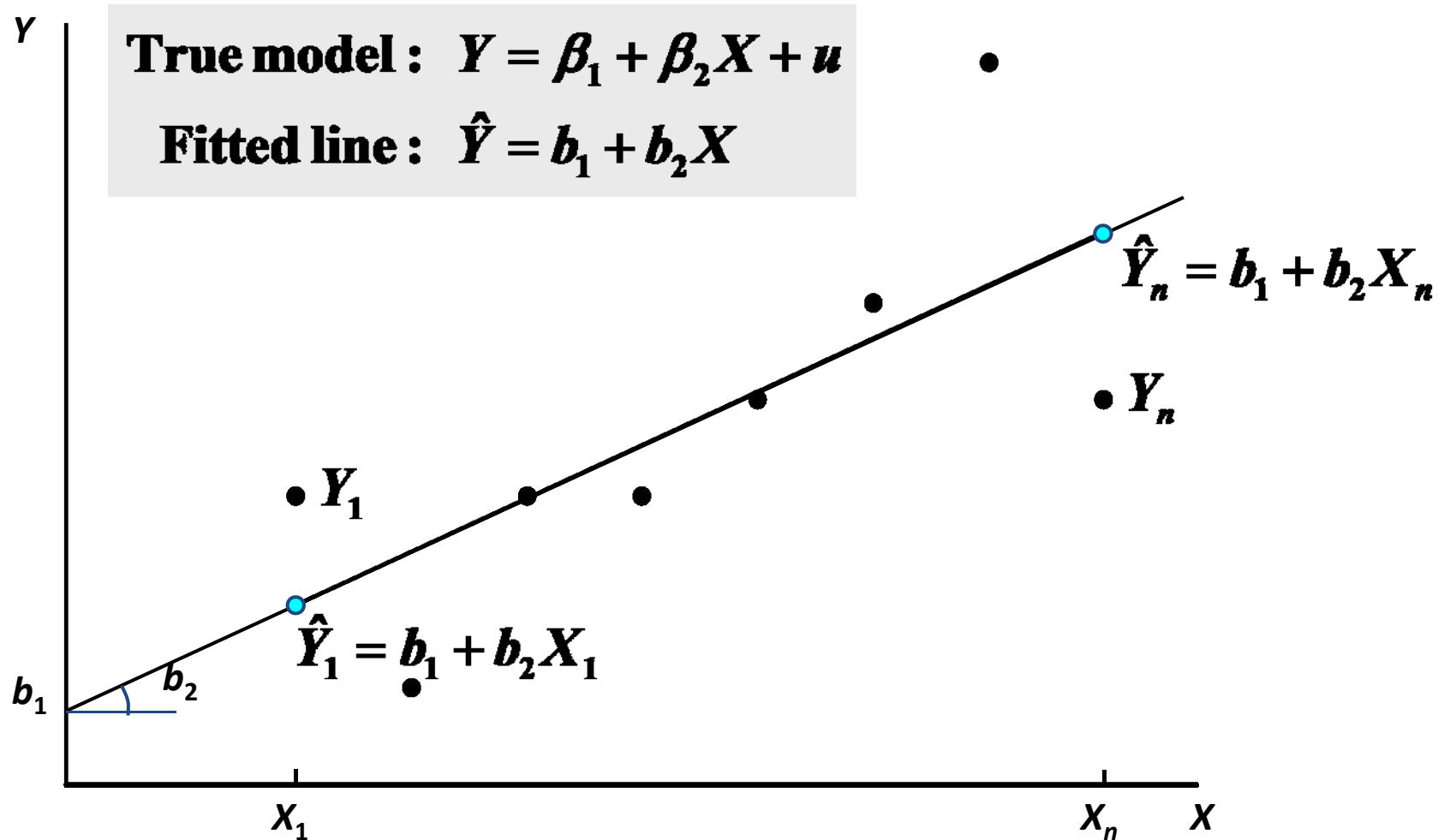
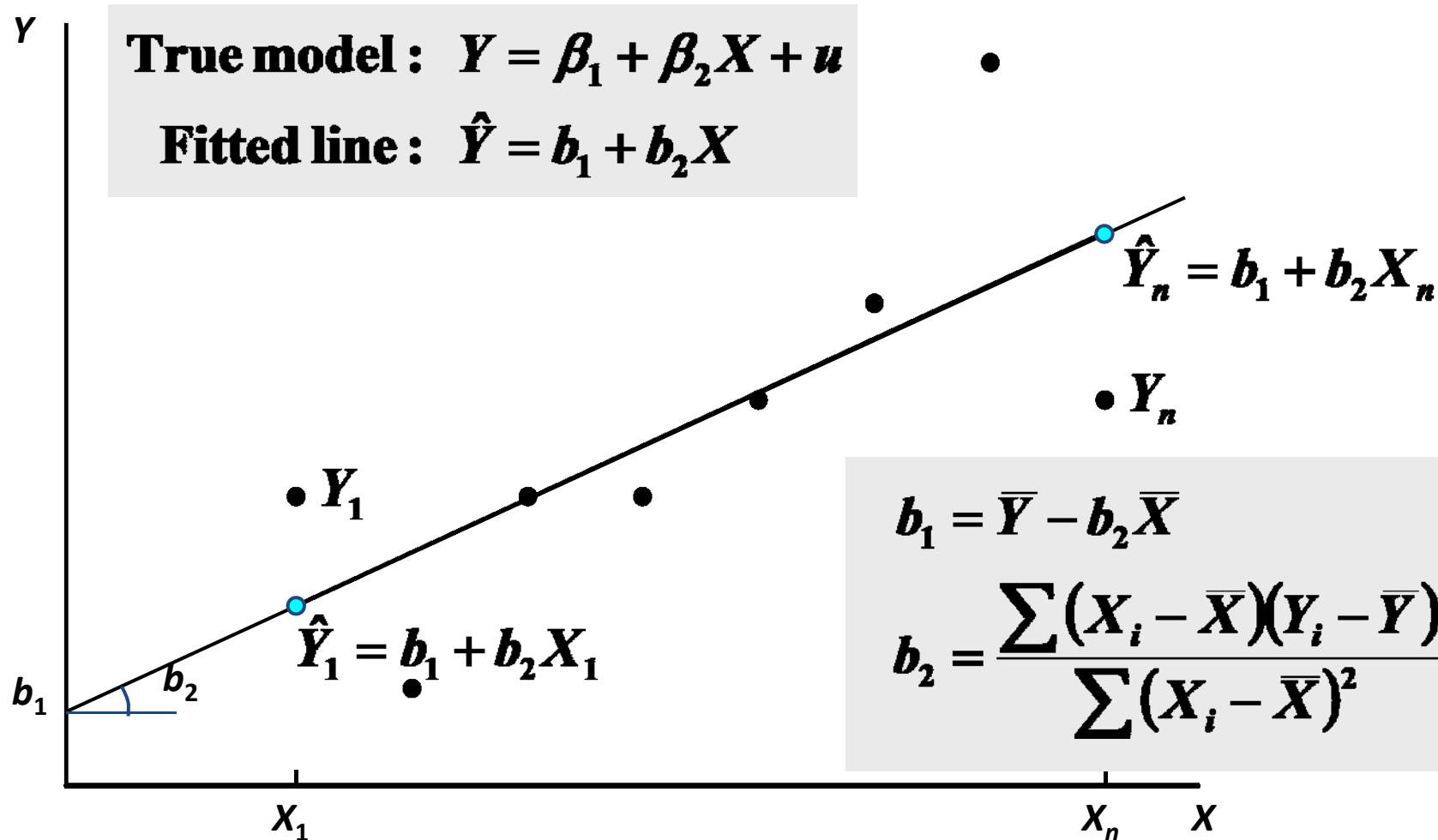


DERIVING LINEAR REGRESSION COEFFICIENTS



The scatter diagram is shown again. We will summarize what we have done. We hypothesized that the true model is as shown, we obtained some data, and we fitted a line.

DERIVING LINEAR REGRESSION COEFFICIENTS



We chose the parameters of the fitted line so as to minimize the sum of the squares of the residuals. As a result, we derived the expressions for b_1 and b_2 .

Cuatro propiedades del modelo MCO:

1. El promedio de los residuos estimados es 0. $\bar{e} = 0$
2. El valor promedio de y es igual al valor promedio de los $\hat{y}$ “ajustados”, es decir, a los que predice el modelo. $\bar{\hat{Y}} = \bar{Y}$
3. La suma de los residuos ponderados por el valor de X correspondiente es 0. $\sum X_i e_i = 0$
4. La suma de los residuos ponderados por el valor de $\hat{Y}$ correspondiente es 0. $\sum \hat{Y}_i e_i = 0$

Bondad de Ajuste

Cuatro propiedades del modelo MCO:

1. El promedio de los residuos estimados es 0. $\bar{e} = 0$

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0$$

$$\bar{\hat{Y}} = \bar{Y}$$

$$\sum X_i e_i = 0$$

$$\sum \hat{Y}_i e_i = 0$$

$$e_i = Y_i - \hat{Y}_i$$

The residual in any observation is given by the difference between the actual and fitted values of Y for that observation.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0$$

$$\bar{\hat{Y}} = \bar{Y}$$

$$\sum X_i e_i = 0$$

$$\sum \hat{Y}_i e_i = 0$$

$$e_i = Y_i - \hat{Y}_i = Y_i - b_1 - b_2 X_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

First substitute for the fitted value.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0$$

$$\bar{\hat{Y}} = \bar{Y}$$

$$\sum X_i e_i = 0$$

$$\sum \hat{Y}_i e_i = 0$$

$$e_i = Y_i - \hat{Y}_i = Y_i - b_1 - b_2 X_i$$

$$\sum e_i = \sum Y_i - nb_1 - b_2 \sum X_i$$

Now sum over all the observations.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0$$

$$\bar{\hat{Y}} = \bar{Y}$$

$$\sum X_i e_i = 0$$

$$\sum \hat{Y}_i e_i = 0$$

$$e_i = Y_i - \hat{Y}_i = Y_i - b_1 - b_2 X_i$$

$$\sum e_i = \sum Y_i - nb_1 - b_2 \sum X_i$$

$$\frac{1}{n} \sum e_i = \frac{1}{n} \sum Y_i - b_1 - b_2 \frac{1}{n} \sum X_i$$

$$\bar{e} = \bar{Y} - b_1 - b_2 \bar{X}$$

Dividing through by n , we obtain the sample mean of the residuals in terms of the sample means of X and Y and the regression coefficients.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0$$

$$\bar{\hat{Y}} = \bar{Y}$$

$$\sum X_i e_i = 0$$

$$\sum \hat{Y}_i e_i = 0$$

$$e_i = Y_i - \hat{Y}_i = Y_i - b_1 - b_2 X_i$$

$$\sum e_i = \sum Y_i - nb_1 - b_2 \sum X_i$$

$$\frac{1}{n} \sum e_i = \frac{1}{n} \sum Y_i - b_1 - b_2 \frac{1}{n} \sum X_i$$

$$\bar{e} = \bar{Y} - b_1 - b_2 \bar{X}$$

$$b_1 = \bar{Y} - b_2 \bar{X}$$

$$= \bar{Y} - (\bar{Y} - b_2 \bar{X}) - b_2 \bar{X} = 0$$

If we substitute for b_1 , the expression collapses to zero.

Cuatro propiedades del modelo MCO:

1. El promedio de los residuos estimados es 0. $\bar{e} = 0$

Lo vemos en Stata

```
sysuse auto  
regress price mpg foreign  
predict double resid, residuals  
sum resid
```

Cuatro propiedades del modelo MCO:

2. El valor promedio de y es igual al valor promedio de los $\hat{y}$ “ajustados”, es decir, a los que predice el modelo. $\bar{\hat{Y}} = \bar{Y}$

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0$$

$$\overline{\hat{Y}} = \bar{Y}$$

$$\sum X_i e_i = 0$$

$$\sum \hat{Y}_i e_i = 0$$

Next we will demonstrate that the mean of the fitted values of Y is equal to the mean of the actual values of Y .

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0$$

$$\overline{\hat{Y}} = \bar{Y}$$

$$\sum X_i e_i = 0$$

$$\sum \hat{Y}_i e_i = 0$$

$$e_i = Y_i - \hat{Y}_i$$

Again, we start with the definition of a residual.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0$$

$$\bar{\hat{Y}} = \bar{Y}$$

$$\sum X_i e_i = 0$$

$$\sum \hat{Y}_i e_i = 0$$

$$e_i = Y_i - \hat{Y}_i$$

$$\sum e_i = \sum Y_i - \sum \hat{Y}_i$$

Sum over all the observations.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0 \quad \bar{\hat{Y}} = \bar{Y} \quad \sum X_i e_i = 0 \quad \sum \hat{Y}_i e_i = 0$$

$$e_i = Y_i - \hat{Y}_i$$

$$\sum e_i = \sum Y_i - \sum \hat{Y}_i$$

$$\frac{1}{n} \sum e_i = \frac{1}{n} \sum Y_i - \frac{1}{n} \sum \hat{Y}_i$$

$$\bar{e} = \bar{Y} - \bar{\hat{Y}}$$

Divide through by n . The terms in the equation are the means of the residuals, actual values of Y , and fitted values of Y , respectively.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0$$

$$\bar{\hat{Y}} = \bar{Y}$$

$$\sum X_i e_i = 0$$

$$\sum \hat{Y}_i e_i = 0$$

$$e_i = Y_i - \hat{Y}_i$$

$$\sum e_i = \sum Y_i - \sum \hat{Y}_i$$

$$\frac{1}{n} \sum e_i = \frac{1}{n} \sum Y_i - \frac{1}{n} \sum \hat{Y}_i$$

$$\bar{e} = \bar{Y} - \bar{\hat{Y}}$$

$$\bar{\hat{Y}} = \bar{Y}$$

We have just shown that the mean of the residuals is zero. Hence the mean of the fitted values is equal to the mean of the actual values.

Cuatro propiedades del modelo MCO:

2. El valor promedio de y es igual al valor promedio de los $\hat{y}$ “ajustados”, es decir, a los que predice el modelo. $\bar{\hat{Y}} = \bar{Y}$

```
predict pricehat, xb
sum price pricehat
graph twoway (scatter price mpg) ///
             (scatter pricehat mpg)
```

Cuatro propiedades del modelo MCO:

3. La suma de los residuos ponderados por el valor de X correspondiente es 0.

$$\sum X_i e_i = 0$$

4. La suma de los residuos ponderados por el valor de Y correspondiente es 0.

$$\sum \hat{Y}_i e_i = 0$$

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0 \qquad \bar{\hat{Y}} = \bar{Y} \qquad \sum X_i e_i = 0 \qquad \sum \hat{Y}_i e_i = 0$$

Next we will demonstrate that the sum of the products of the values of X and the residuals is zero.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0 \qquad \bar{\hat{Y}} = \bar{Y} \qquad \sum X_i e_i = 0 \qquad \sum \hat{Y}_i e_i = 0$$

$$\sum X_i e_i = \sum X_i (Y_i - b_1 - b_2 X_i)$$

We start by replacing the residual with its expression in terms of Y and X .

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0 \qquad \bar{\hat{Y}} = \bar{Y} \qquad \sum X_i e_i = 0 \qquad \sum \hat{Y}_i e_i = 0$$

$$\begin{aligned} \sum X_i e_i &= \sum X_i (Y_i - b_1 - b_2 X_i) \\ &= \sum X_i Y_i - b_1 \sum X_i - b_2 \sum X_i^2 \end{aligned}$$

We expand the expression.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0 \qquad \bar{\hat{Y}} = \bar{Y} \qquad \sum X_i e_i = 0 \qquad \sum \hat{Y}_i e_i = 0$$

$$\begin{aligned} \sum X_i e_i &= \sum X_i (Y_i - b_1 - b_2 X_i) \\ &= \sum X_i Y_i - b_1 \sum X_i - b_2 \sum X_i^2 = 0 \end{aligned}$$

The expression is equal to zero. One way of demonstrating this would be to substitute for b_1 and b_2 and show that all the terms cancel out.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0 \quad \bar{\hat{Y}} = \bar{Y} \quad \sum X_i e_i = 0 \quad \sum \hat{Y}_i e_i = 0$$

$$\begin{aligned} \sum X_i e_i &= \sum X_i (Y_i - b_1 - b_2 X_i) \\ &= \sum X_i Y_i - b_1 \sum X_i - b_2 \sum X_i^2 = 0 \end{aligned}$$

$$\frac{\partial RSS}{\partial b_2} = 0 \Rightarrow 2b_2 \sum X_i^2 - 2 \sum X_i Y_i + 2b_1 \sum X_i = 0$$

A neater way is to recall the first order condition for b_2 when deriving the regression coefficients. You can see that it is exactly what we need.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0 \qquad \bar{\hat{Y}} = \bar{Y} \qquad \sum X_i e_i = 0 \qquad \sum \hat{Y}_i e_i = 0$$

Finally we will demonstrate that the sum of the products of the fitted values of Y and the residuals is zero.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0 \quad \bar{\hat{Y}} = \bar{Y} \quad \sum X_i e_i = 0 \quad \sum \hat{Y}_i e_i = 0$$

$$\sum \hat{Y}_i e_i = \sum (b_1 + b_2 X_i) e_i$$

We start by substituting for the fitted value of Y .

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0 \qquad \bar{\hat{Y}} = \bar{Y} \qquad \sum X_i e_i = 0 \qquad \sum \hat{Y}_i e_i = 0$$

$$\begin{aligned} \sum \hat{Y}_i e_i &= \sum (b_1 + b_2 X_i) e_i \\ &= \sum b_1 e_i + \sum b_2 X_i e_i \\ &= b_1 n \bar{e} + b_2 \sum X_i e_i \end{aligned}$$

$\sum e_i = n \bar{e}$ $\uparrow$

We expand and rearrange.

GOODNESS OF FIT

Four useful results:

$$\bar{e} = 0 \quad \bar{\hat{Y}} = \bar{Y} \quad \sum X_i e_i = 0 \quad \sum \hat{Y}_i e_i = 0$$

$$\begin{aligned} \sum \hat{Y}_i e_i &= \sum (b_1 + b_2 X_i) e_i \\ &= \sum b_1 e_i + \sum b_2 X_i e_i \\ &= b_1 n \bar{e} + b_2 \sum X_i e_i = 0 \end{aligned}$$

The expression is equal to zero, given the first and third useful results.

Cuatro propiedades del modelo MCO:

3. La suma de los residuos ponderados por el valor de X correspondiente es 0.

$$\sum X_i e_i = 0$$

4. La suma de los residuos ponderados por el valor de Y correspondiente es 0.

$$\sum \hat{Y}_i e_i = 0$$

Tarea: Mostrar estos resultados en Stata

Bondad de Ajuste

Estas 4 propiedades nos permiten demostrar un resultado que es importante para comprender el ajuste de nuestros modelos econométricos.

$$\text{TSS} = \text{ESS} + \text{RSS}$$

La suma de las desviaciones cuadradas de la variable a explicar (TSS) se igualan a la suma de:

- i) La suma de las desviaciones explicadas por el modelo al cuadrado (ESS)**
- ii) La suma de los residuos al cuadrado del modelo (RSS)**

Bondad de Ajuste

$$\sum (y_i - \bar{Y})^2$$

We now come to the discussion of goodness of fit. One measure of the variation in Y is the sum of its squared deviations around **its sample mean**, often described as the Total Sum of Squares, TSS .

GOODNESS OF FIT

$$e_i = Y_i - \hat{Y}_i \Rightarrow Y_i = \hat{Y}_i + e_i$$

$$\sum (Y_i - \bar{Y})^2$$

We will decompose *TSS* using the fact that the actual value of *Y* in any observations is equal to the sum of its fitted value and the residual.

GOODNESS OF FIT

$$e_i = Y_i - \hat{Y}_i \Rightarrow Y_i = \hat{Y}_i + e_i$$

$$\sum (Y_i - \bar{Y})^2 = \sum ([\hat{Y}_i + e_i] - [\bar{\hat{Y}} + \bar{e}])^2$$

We substitute for Y_i .

GOODNESS OF FIT

$$e_i = Y_i - \hat{Y}_i \Rightarrow Y_i = \hat{Y}_i + e_i$$

$$\sum (Y_i - \bar{Y})^2 = \sum ([\hat{Y}_i + e_i] - [\bar{\hat{Y}} + \bar{e}])^2$$

$\nearrow \quad \nwarrow$
 $\bar{\hat{Y}} = \bar{Y} \quad \bar{e} = 0$

From the useful results, the mean of the fitted values of Y is equal to the mean of the actual values. Also, the mean of the residuals is zero.

GOODNESS OF FIT

$$e_i = Y_i - \hat{Y}_i \Rightarrow Y_i = \hat{Y}_i + e_i$$

$$\sum (Y_i - \bar{Y})^2 = \sum ([\hat{Y}_i + e_i] - [\bar{\hat{Y}} + \bar{e}])^2 = \sum ([\hat{Y}_i - \bar{Y}] + e_i)^2$$

$\nearrow \quad \nwarrow$
 $\bar{\hat{Y}} = \bar{Y} \quad \bar{e} = 0$

Hence we can simplify the expression as shown.

GOODNESS OF FIT

$$e_i = Y_i - \hat{Y}_i \Rightarrow Y_i = \hat{Y}_i + e_i$$

$$\begin{aligned}\sum (Y_i - \bar{Y})^2 &= \sum ([\hat{Y}_i + e_i] - [\bar{\hat{Y}} + \bar{e}])^2 = \sum ([\hat{Y}_i - \bar{Y}] + e_i)^2 \\ &= \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 + 2\sum ([\hat{Y}_i - \bar{Y}]e_i)\end{aligned}$$

We expand the squared terms on the right side of the equation.

GOODNESS OF FIT

$$e_i = Y_i - \hat{Y}_i \Rightarrow Y_i = \hat{Y}_i + e_i$$

$$\begin{aligned}\sum (Y_i - \bar{Y})^2 &= \sum ([\hat{Y}_i + e_i] - [\bar{\hat{Y}} + \bar{e}])^2 = \sum ([\hat{Y}_i - \bar{Y}] + e_i)^2 \\&= \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 + 2\sum ([\hat{Y}_i - \bar{Y}]e_i) \\&= \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 + 2\sum \hat{Y}_i e_i - 2\bar{Y} \sum e_i\end{aligned}$$

We expand the third term on the right side of the equation.

GOODNESS OF FIT

$$e_i = Y_i - \hat{Y}_i \Rightarrow Y_i = \hat{Y}_i + e_i$$

$$\sum (Y_i - \bar{Y})^2 = \sum ([\hat{Y}_i + e_i] - [\bar{\hat{Y}} + \bar{e}])^2 = \sum ([\hat{Y}_i - \bar{Y}] + e_i)^2$$

$$= \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 + 2 \sum ([\hat{Y}_i - \bar{Y}] e_i)$$

$$= \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 + 2 \sum \hat{Y}_i e_i - 2 \bar{Y} \sum e_i$$

$$\begin{array}{ccc} \uparrow & & \uparrow \\ \sum \hat{Y}_i e_i = 0 & & \bar{e} = 0, \text{ so} \\ & & \sum e_i = 0 \end{array}$$

The last two terms are both zero, given the first and fourth useful results.

GOODNESS OF FIT

$$e_i = Y_i - \hat{Y}_i \Rightarrow Y_i = \hat{Y}_i + e_i$$

$$\sum (Y_i - \bar{Y})^2 = \sum ([\hat{Y}_i + e_i] - [\bar{\hat{Y}} + \bar{e}])^2 = \sum ([\hat{Y}_i - \bar{Y}] + e_i)^2$$

$$= \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 + 2 \sum ([\hat{Y}_i - \bar{Y}] e_i)$$

$$= \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 + 2 \sum \hat{Y}_i e_i - 2 \bar{Y} \sum e_i$$

$$\sum (Y_i - \bar{Y})^2 = \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 \quad TSS = ESS + RSS$$

Thus we have shown that *TSS*, the total sum of squares of *Y* can be decomposed into *ESS*, the ‘explained’ sum of squares, and *RSS*, the residual (‘unexplained’) sum of squares.

GOODNESS OF FIT

$$e_i = Y_i - \hat{Y}_i \Rightarrow Y_i = \hat{Y}_i + e_i$$

$$\sum (Y_i - \bar{Y})^2 = \sum ([\hat{Y}_i + e_i] - [\bar{\hat{Y}} + \bar{e}])^2 = \sum ([\hat{Y}_i - \bar{Y}] + e_i)^2$$

$$= \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 + 2 \sum ([\hat{Y}_i - \bar{Y}] e_i)$$

$$= \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 + 2 \sum \hat{Y}_i e_i - 2 \bar{Y} \sum e_i$$

$$\sum (Y_i - \bar{Y})^2 = \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 \quad TSS = ESS + RSS$$

The words *explained* and *unexplained* were put in quotation marks because the explanation may in fact be false. Y might really depend on some other variable Z , and X might be acting as a proxy for Z . It would be safer to use the expression *apparently explained* instead of *explained*.

Bondad de Ajuste

La propiedad $TSS=ESS+RSS$ sugiere que una buena medida de bondad de ajuste es:

$$R^2 = \frac{ESS}{TSS} = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2}$$

Le llamamos R cuadrado de la regresión

GOODNESS OF FIT

$$R^2 = \frac{ESS}{TSS} = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2}$$

$$\sum (Y_i - \bar{Y})^2 = \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 \quad TSS = ESS + RSS$$

The main criterion of goodness of fit, formally described as the coefficient of determination, but usually referred to as R^2 , is defined to be the ratio of ESS to TSS , that is, the proportion of the variance of Y explained by the regression equation.

GOODNESS OF FIT

$$R^2 = \frac{ESS}{TSS} = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2}$$

$$\sum (Y_i - \bar{Y})^2 = \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 \quad TSS = ESS + RSS$$

Obviously we would like to locate the regression line so as to make the goodness of fit as high as possible, according to this criterion. Does this objective clash with our use of the least squares principle to determine b_1 and b_2 ?

GOODNESS OF FIT

$$R^2 = \frac{ESS}{TSS} = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2}$$

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{\sum e_i^2}{\sum (Y_i - \bar{Y})^2}$$

$$\sum (Y_i - \bar{Y})^2 = \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 \quad TSS = ESS + RSS$$

Fortunately, there is no clash. To see this, rewrite the expression for R^2 in term of RSS as shown.

GOODNESS OF FIT

$$R^2 = \frac{ESS}{TSS} = \frac{\sum (\hat{Y}_i - \bar{Y})^2}{\sum (Y_i - \bar{Y})^2}$$

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{\sum e_i^2}{\sum (Y_i - \bar{Y})^2}$$

$$\sum (Y_i - \bar{Y})^2 = \sum (\hat{Y}_i - \bar{Y})^2 + \sum e_i^2 \quad TSS = ESS + RSS$$

The OLS regression coefficients are chosen in such a way as to minimize the sum of the squares of the residuals. Thus it automatically follows that they maximize R^2 .

Bondad de Ajuste

Calculamos los TSS, ESS y RSS en Stata

Verificar que sucede con el R^2 de la regresión al aumentar variables.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Suppose that the units of measurement of Y or X are changed. How will this affect the regression results? Intuitively, we would anticipate that nothing of substance will be changed, and this is correct.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

We will demonstrate this for the estimates of the regression coefficients in this section, and we will trace the implications for the rest of the regression output in due course. We begin by supposing that the true and fitted models are as shown above.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$Y_i^* = \lambda_1 + \lambda_2 Y_i$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

We now suppose that the units of measurement of Y are changed, with the new measure, Y^* , being a linear function of the old one.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$Y_i^* = \lambda_1 + \lambda_2 Y_i$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Typically, a change of measurement involves a simple multiplicative scaling, such as when we convert pounds into grams.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$Y_i^* = \lambda_1 + \lambda_2 Y_i$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

However, one occasionally encounters a full linear transformation. Conversion of temperatures from degrees Celsius to degrees Fahrenheit is an example.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$Y_i^* = \lambda_1 + \lambda_2 Y_i$$

$$b_2^* = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i^* - \bar{Y}^*)}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Regressing Y^* on X , the new slope coefficient b_2^* is as shown.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$Y_i^* = \lambda_1 + \lambda_2 Y_i$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$b_2^* = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i^* - \bar{Y}^*)}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})([\lambda_1 + \lambda_2 Y_i] - [\lambda_1 + \lambda_2 \bar{Y}])}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

We substitute for Y^* from the linear relationship defining it.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$Y_i^* = \lambda_1 + \lambda_2 Y_i$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\begin{aligned} b_2^* &= \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i^* - \bar{Y}^*)}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})([\lambda_1 + \lambda_2 Y_i] - [\lambda_1 + \lambda_2 \bar{Y}])}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ &= \frac{\sum_{i=1}^n (X_i - \bar{X})(\lambda_2 Y_i - \lambda_2 \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\lambda_2 \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} \end{aligned}$$

The λ_1 terms cancel. λ_2 is a common factor in the numerator.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$Y_i^* = \lambda_1 + \lambda_2 Y_i$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\begin{aligned} b_2^* &= \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i^* - \bar{Y}^*)}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})([\lambda_1 + \lambda_2 Y_i] - [\lambda_1 + \lambda_2 \bar{Y}])}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ &= \frac{\sum_{i=1}^n (X_i - \bar{X})(\lambda_2 Y_i - \lambda_2 \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\lambda_2 \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \lambda_2 b_2 \end{aligned}$$

We find that the new slope coefficient is equal to the original one, multiplied by λ_2 .

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$Y_i^* = \lambda_1 + \lambda_2 Y_i$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\begin{aligned} b_2^* &= \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i^* - \bar{Y}^*)}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})([\lambda_1 + \lambda_2 Y_i] - [\lambda_1 + \lambda_2 \bar{Y}])}{\sum_{i=1}^n (X_i - \bar{X})^2} \\ &= \frac{\sum_{i=1}^n (X_i - \bar{X})(\lambda_2 Y_i - \lambda_2 \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\lambda_2 \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \lambda_2 b_2 \end{aligned}$$

This is logical. A unit change in Y is the same as a change of λ_2 units in Y^* . According to the regression equation, a unit change in X leads to a change of b_2 units in Y , so it should lead to a change of $\lambda_2 b_2$ units in Y^* .

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$\hat{Y}_i = b_1 + b_2 X_i$$

$$Y_i^* = \lambda_1 + \lambda_2 Y_i$$

$$b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Ejemplo en Stata, pasar las unidades de dolares a pesos.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$X_i^* = X_i - \bar{X}$$

However, we will consider a special case of a change in the measurement of X . Often the intercept in a regression equation has no sensible interpretation because $X = 0$ is distant from the data range.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$X_i^* = X_i - \bar{X}$$

The earnings function illustrated in the previous slideshow is an example, with the intercept actually being negative.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$X_i^* = X_i - \bar{X}$$

Sometimes it is useful to deal with the problem by defining X^* as the deviation of X about its sample mean.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$X_i^* = X_i - \bar{X}$$

$$\sum_{i=1}^n X_i^* = \sum_{i=1}^n (X_i - \bar{X}) = \sum_{i=1}^n X_i - n\bar{X} = n\bar{X} - n\bar{X} = 0$$

$$\bar{X}^* = \frac{1}{n} \sum_{i=1}^n X_i^* = 0$$

Note that, by definition, the sample sum of X_i^* is 0, and hence the mean value of X^* is zero..

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$X_i^* = X_i - \bar{X}$$

$$\bar{X}^* = \frac{1}{n} \sum_{i=1}^n X_i^* = 0$$

$$\hat{Y}_i = b_1^* + b_2^* X_i^*$$

$$b_2^* = \frac{\sum_{i=1}^n (X_i^* - \bar{X}^*)(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i^* - \bar{X}^*)^2} = \frac{\sum_{i=1}^n X_i^* (Y_i - \bar{Y})}{\sum_{i=1}^n X_i^{*2}} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = b_2$$

If we now regress Y on X^* instead of X , the slope coefficient will not be affected.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$X_i^* = X_i - \bar{X}$$

$$\bar{X}^* = \frac{1}{n} \sum_{i=1}^n X_i^* = 0$$

$$\hat{Y}_i = b_1^* + b_2^* X_i^*$$

$$b_2^* = \frac{\sum_{i=1}^n (X_i^* - \bar{X}^*)(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i^* - \bar{X}^*)^2} = \frac{\sum_{i=1}^n X_i^* (Y_i - \bar{Y})}{\sum_{i=1}^n X_i^{*2}} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = b_2$$

$$b_1^* = \bar{Y} - b_2^* \bar{X}^* = \bar{Y}$$

The intercept will now be the fitted value of Y at the sample mean of X . This is the sample mean of Y and it may be more informative for analytical purposes.

CHANGES IN THE UNITS OF MEASUREMENT

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

$$X_i^* = X_i - \bar{X}$$

$$b_1^* = \bar{Y} - b_2^* \bar{X}^* = \bar{Y}$$

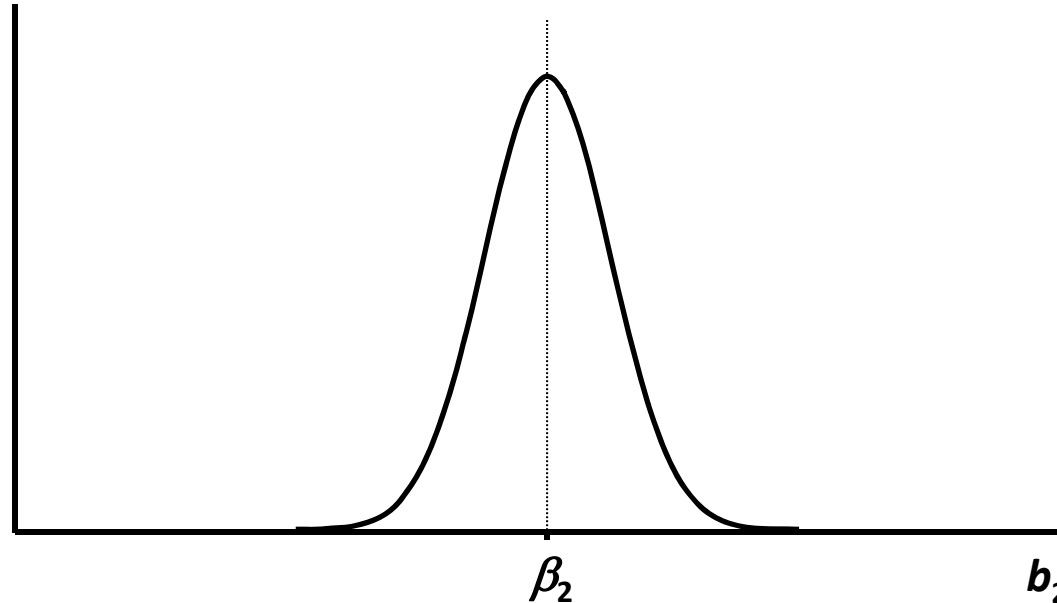
Ejemplo en Stata:

Extraer la media al peso de los automoviles.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

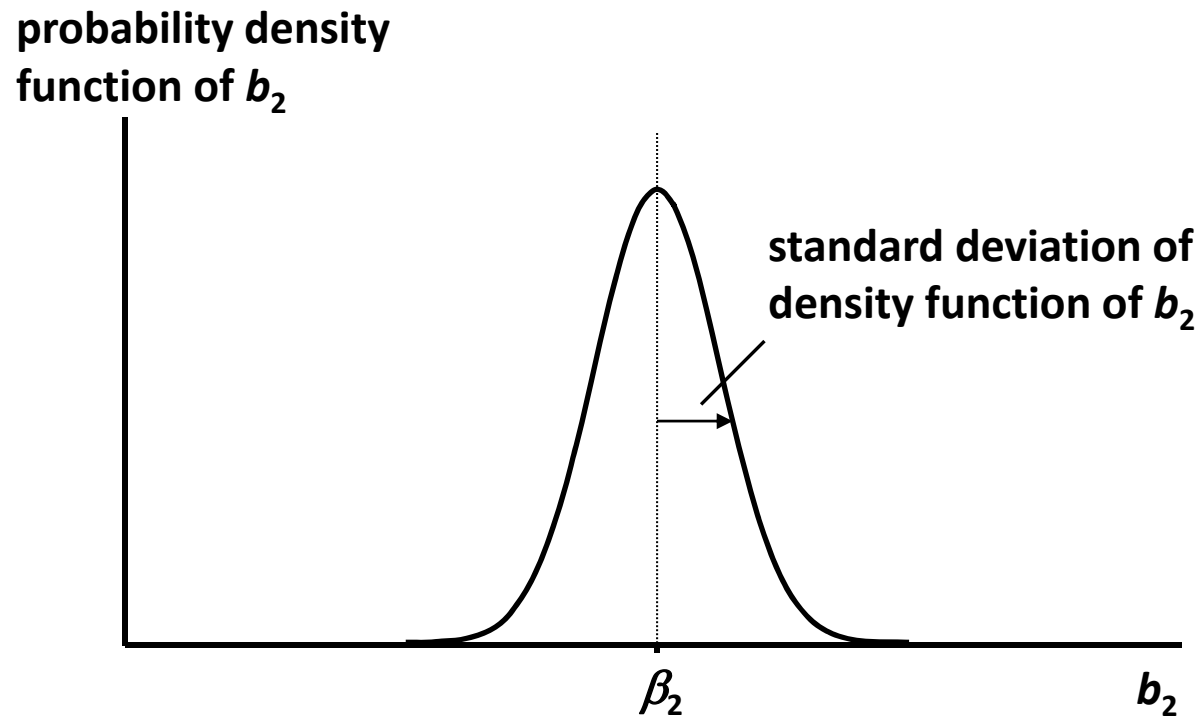
probability density
function of b_2



We have seen that the regression coefficients b_1 and b_2 are random variables. They provide point estimates of β_1 and β_2 , respectively. In the last sequence we demonstrated that these point estimates are unbiased.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$



In this sequence we will see that we can also obtain estimates of the standard deviations of the distributions. These will give some idea of their likely reliability and will provide a basis for tests of hypotheses.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\}$$

$$\sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2}$$

Expressions (which will not be derived) for the variances of their distributions are shown above. See Box 2.3 in the text for a proof of the expression for the variance of b_2 .

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

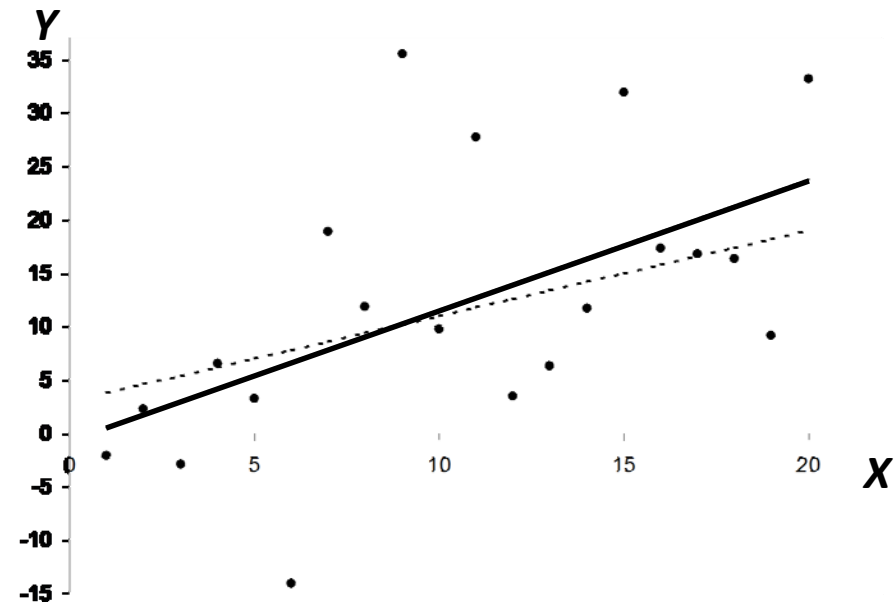
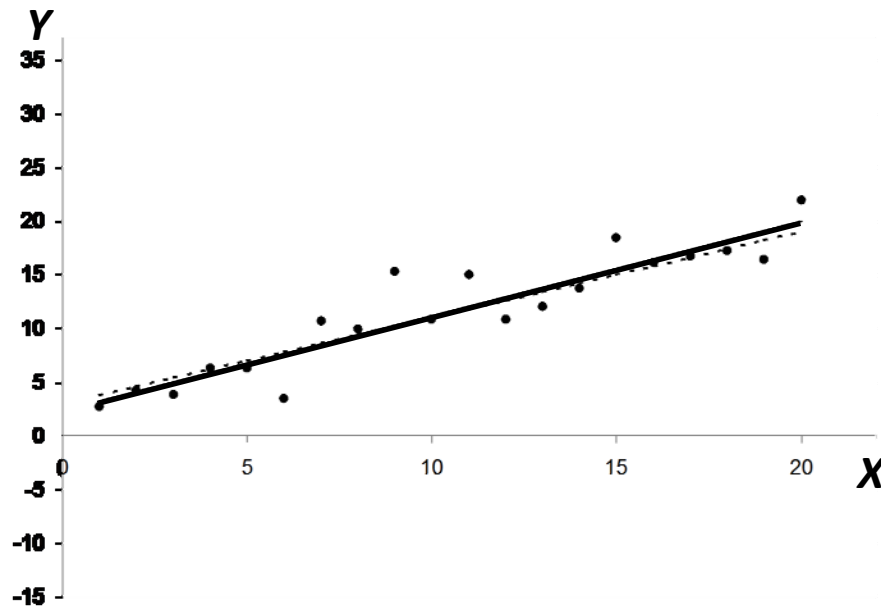
$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\}$$

$$\sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2}$$

We will focus on the implications of the expression for the variance of b_2 . Looking at the numerator, we see that the variance of b_2 is proportional to σ_u^2 . This is as we would expect. The more noise there is in the model, the less precise will be our estimates.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

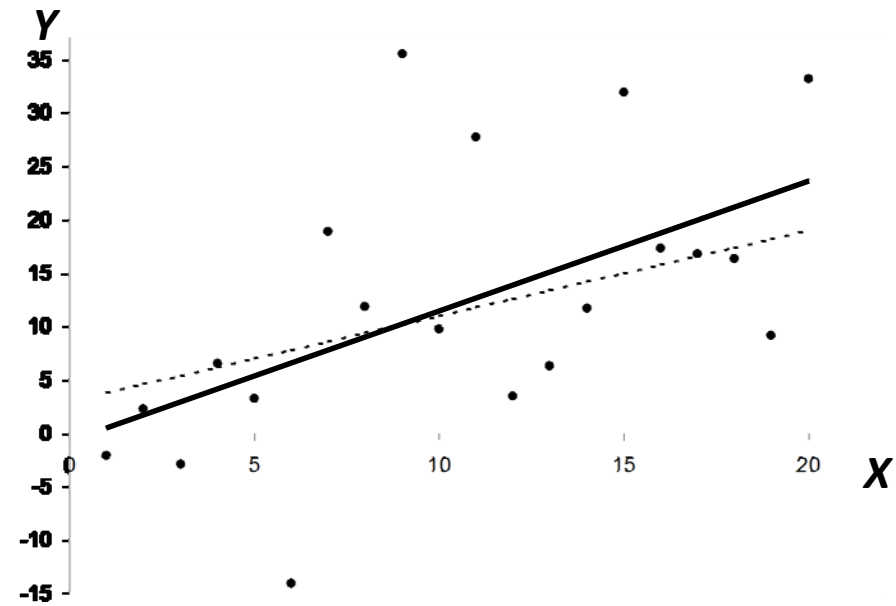
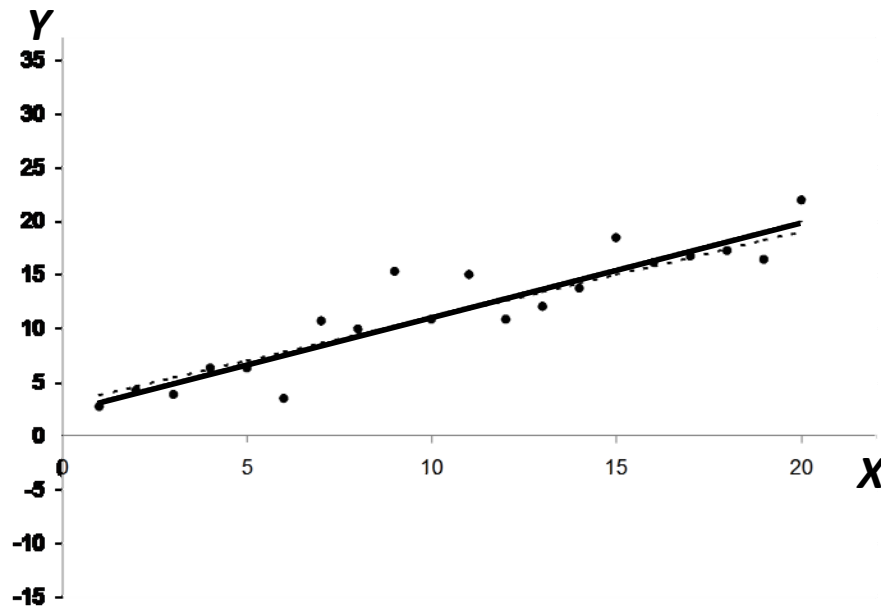


$$Y = 3.0 + 0.8X$$

This is illustrated by the diagrams above. The nonstochastic component of the relationship, $Y = 3.0 + 0.8X$, represented by the dotted line, is the same in both diagrams.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

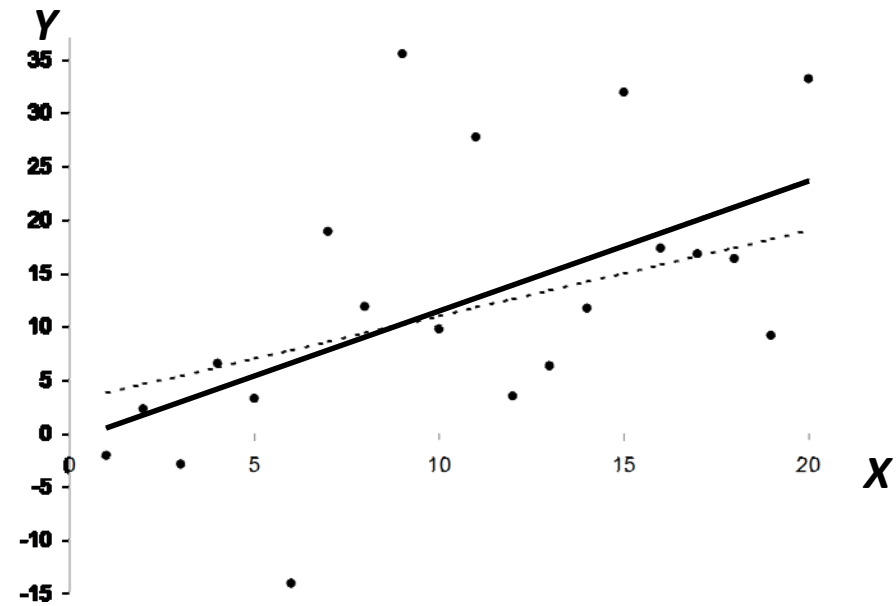
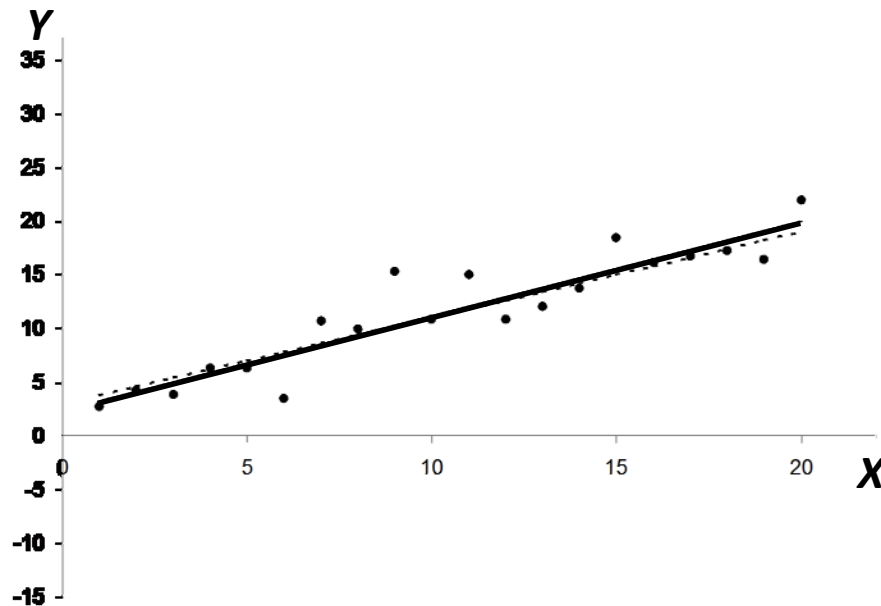


$$Y = 3.0 + 0.8X$$

The values of X are the same, and the same random numbers have been used to generate the values of the disturbance term in the 20 observations.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$



$$Y = 3.0 + 0.8X$$

However, in the right-hand diagram the random numbers have been multiplied by a factor of 5. As a consequence, the regression line, the solid line, is a much poorer approximation to the nonstochastic relationship.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\}$$

$$\sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2}$$

Looking at the denominator, the larger is the sum of the squared deviations of X , the smaller is the variance of b_2 .

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\}$$

$$\sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$\text{MSD}(X) = \frac{1}{n} \sum (X_i - \bar{X})^2$$

However the size of the sum of the squared deviations depends on two factors: the number of observations, and the size of the deviations of X_i about its sample mean. To discriminate between them, it is convenient to define the mean square deviation of X , $\text{MSD}(X)$.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\}$$

$$\sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$\text{MSD}(X) = \frac{1}{n} \sum (X_i - \bar{X})^2$$

From the expression as rewritten, it can be seen that the variance of b_2 is inversely proportional to n , the number of observations in the sample, controlling for $\text{MSD}(X)$. The more information you have, the more accurate your estimates are likely to be.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\}$$

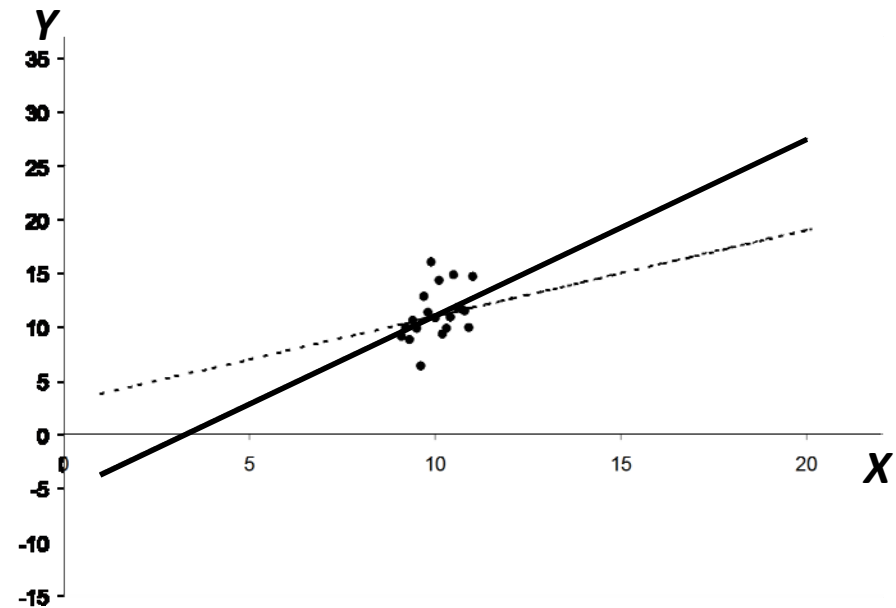
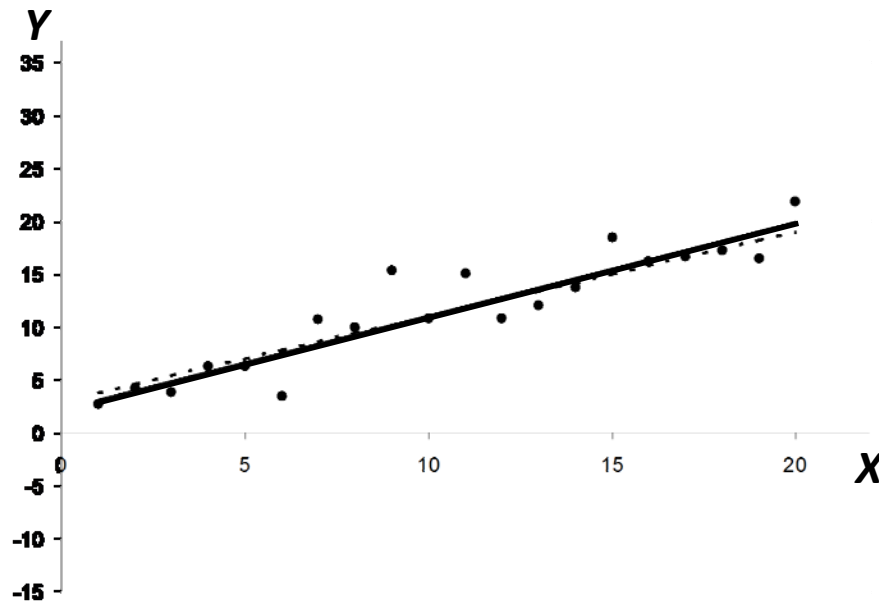
$$\sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$\text{MSD}(X) = \frac{1}{n} \sum (X_i - \bar{X})^2$$

A third implication of the expression is that the variance is inversely proportional to the mean square deviation of X . What is the reason for this?

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

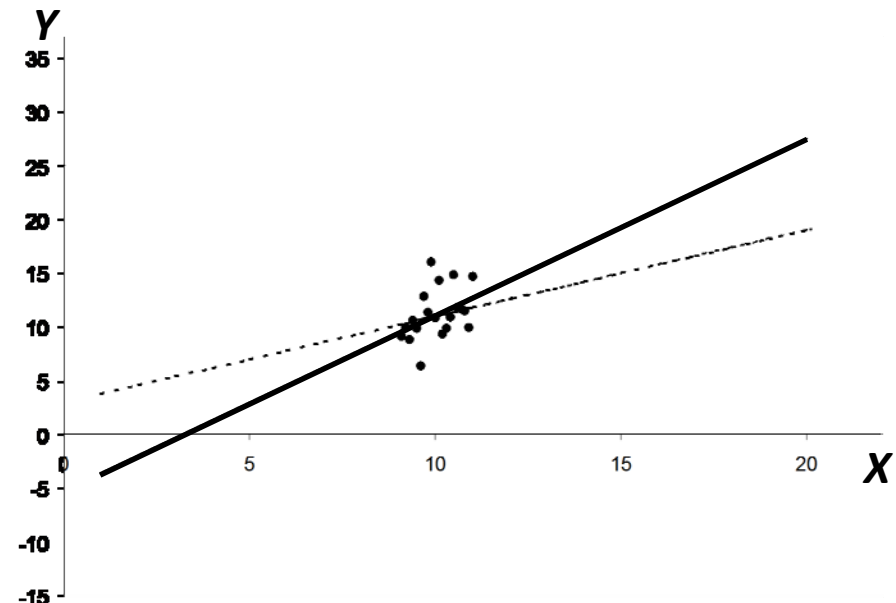
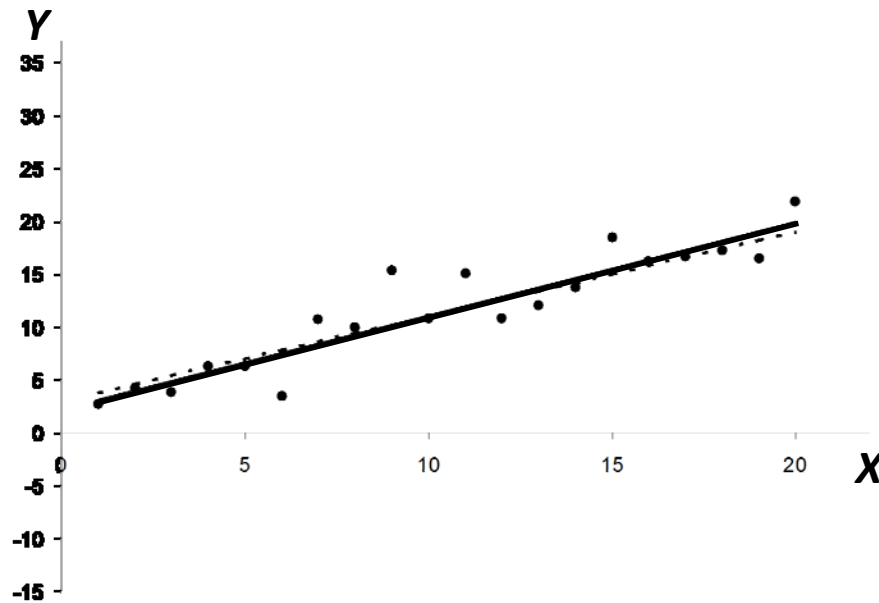


$$Y = 3.0 + 0.8X$$

In the diagrams above, the nonstochastic component of the relationship is the same and the same random numbers have been used for the 20 values of the disturbance term.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

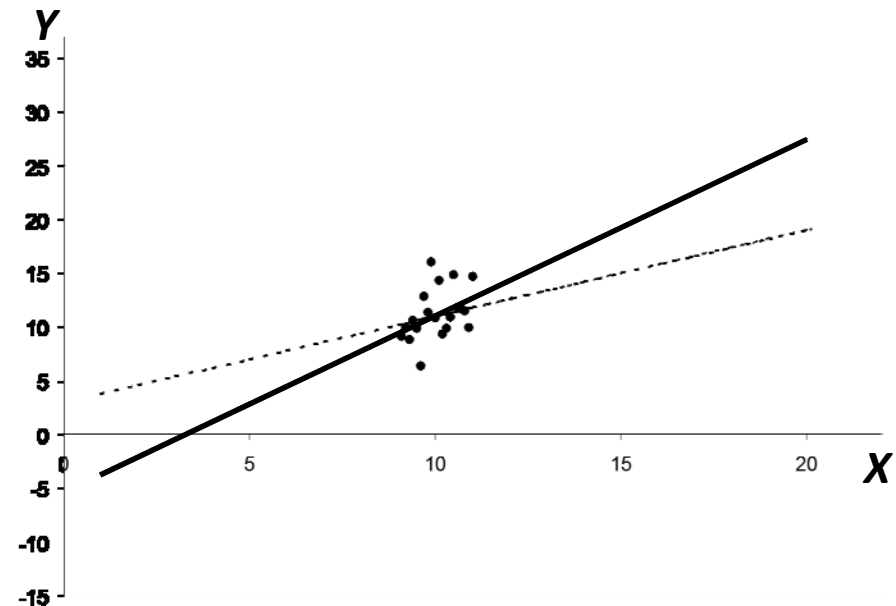
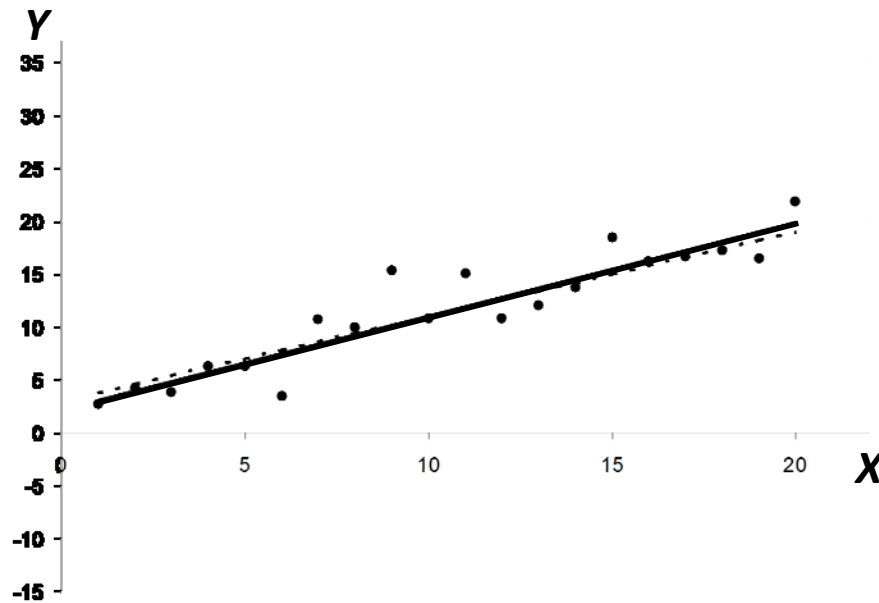


$$Y = 3.0 + 0.8X$$

However, $\text{MSD}(X)$ is much smaller in the right-hand diagram because the values of X are much closer together.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$



$$Y = 3.0 + 0.8X$$

Hence in that diagram the position of the regression line is more sensitive to the values of the disturbance term, and as a consequence the regression line is likely to be relatively inaccurate.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

Of course, as can be seen from the variance expressions, it is really the ratio of the $\text{MSD}(X)$ to the variance of u which is important, rather than the absolute size of either.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

We cannot calculate the variances exactly because we do not know the variance of the disturbance term. However, we can derive an estimator of σ_u^2 from the residuals.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

Clearly the scatter of the residuals around the regression line will reflect the unseen scatter of u about the line $Y_i = \beta_1 + b_2 X_i$, although in general the residual and the value of the disturbance term in any given observation are not equal to one another.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$\text{MSD}(e) = \frac{1}{n} \sum (e_i - \bar{e})^2 = \frac{1}{n} \sum e_i^2$$

One measure of the scatter of the residuals is their mean square error, $\text{MSD}(e)$, defined as shown. (Remember that the mean of the OLS residuals is equal to zero). Intuitively this should provide a guide to the variance of u .

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$\text{MSD}(e) = \frac{1}{n} \sum (e_i - \bar{e})^2 = \frac{1}{n} \sum e_i^2$$

Before going any further, ask yourself the following question. Which line is likely to be closer to the points representing the sample of observations on X and Y , the true line

$Y = \beta_1 + \beta_2 X$ or the regression line $Y = \hat{b}_1 + b_2 X$?

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$\text{MSD}(e) = \frac{1}{n} \sum (e_i - \bar{e})^2 = \frac{1}{n} \sum e_i^2$$

The answer is the regression line, because by definition it is drawn in such a way as to minimize the sum of the squares of the distances between it and the observations.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$\text{MSD}(e) = \frac{1}{n} \sum (e_i - \bar{e})^2 = \frac{1}{n} \sum e_i^2$$

Hence the spread of the residuals will tend to be smaller than the spread of the values of u , and $\text{MSD}(e)$ will tend to underestimate σ_u^2 .

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$\text{MSD}(e) = \frac{1}{n} \sum (e_i - \bar{e})^2 = \frac{1}{n} \sum e_i^2$$

$$E(\text{MSD}(e)) = \frac{n-2}{n} \sigma_u^2$$

Indeed, it can be shown that the expected value of $\text{MSD}(e)$, when there is just one explanatory variable, is given by the expression above.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$s_u^2 = \frac{n}{n-2} \text{MSD}(e) = \frac{n}{n-2} \frac{1}{n} \sum e_i^2 = \frac{1}{n-2} \sum e_i^2$$

However, it follows that we can obtain an unbiased estimator of σ_u^2 by multiplying $\text{MSD}(e)$ by $n / (n - 2)$. We will denote this s_u^2 .

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$s_u^2 = \frac{n}{n-2} \text{MSD}(e) = \frac{n}{n-2} \frac{1}{n} \sum e_i^2 = \frac{1}{n-2} \sum e_i^2$$

$$\text{s.e.}(b_1) = s_u \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2}} \quad \text{s.e.}(b_2) = \sqrt{\frac{s_u^2}{\sum (X_i - \bar{X})^2}}$$

We can then obtain estimates of the standard deviations of the distributions of b_1 and b_2 by substituting s_u^2 for σ_u^2 in the variance expressions and taking the square roots.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

$$\sigma_{b_1}^2 = \sigma_u^2 \left\{ \frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right\} \quad \sigma_{b_2}^2 = \frac{\sigma_u^2}{\sum (X_i - \bar{X})^2} = \frac{\sigma_u^2}{n \text{MSD}(X)}$$

$$s_u^2 = \frac{n}{n-2} \text{MSD}(e) = \frac{n}{n-2} \frac{1}{n} \sum e_i^2 = \frac{1}{n-2} \sum e_i^2$$

$$\text{s.e.}(b_1) = s_u \sqrt{\frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2}} \quad \text{s.e.}(b_2) = \sqrt{\frac{s_u^2}{\sum (X_i - \bar{X})^2}}$$

These are described as the standard errors of b_1 and b_2 , 'estimates of the standard deviations' being a bit of a mouthful.

PRECISION OF THE REGRESSION COEFFICIENTS

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540		
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000
Total	112010.231	539	207.811189	R-squared	=	0.1725
				Adj R-squared	=	0.1710
				Root MSE	=	13.126

EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

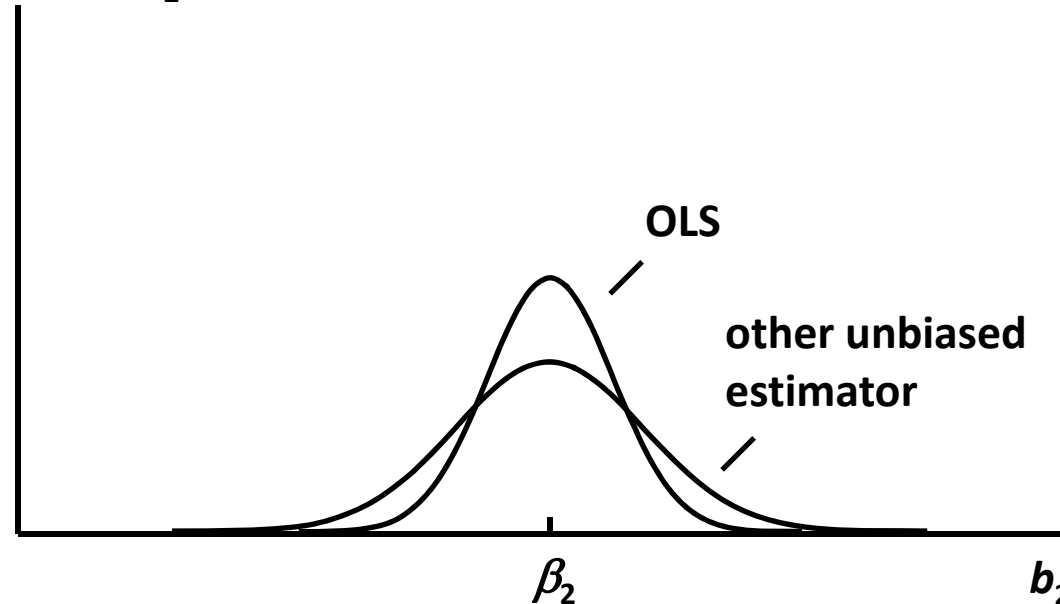
The standard errors of the coefficients always appear as part of the output of a regression. Here is the regression of hourly earnings on years of schooling discussed in a previous slideshow. The standard errors appear in a column to the right of the coefficients.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

Efficiency

probability density
function of b_2

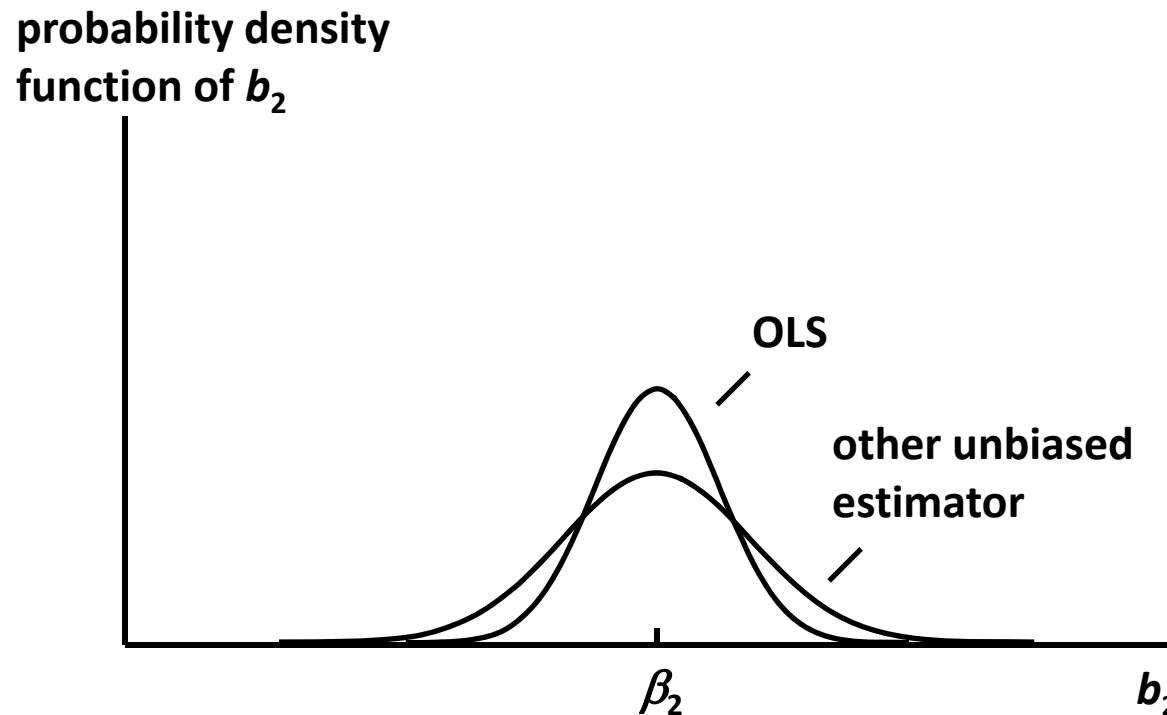


The Gauss–Markov theorem states that, provided that the regression model assumptions are valid, the OLS estimators are BLUE: best (most efficient) linear (functions of the values of Y) unbiased estimators of the parameters.

PRECISION OF THE REGRESSION COEFFICIENTS

Simple regression model: $Y = \beta_1 + \beta_2 X + u$

Efficiency



The proof of the theorem is not difficult but it is not high priority and we will take it on trust. See Section 2.7 of the text for a proof for the simple regression model.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

REVIEW CHAPTER

A random variable X is distributed with unknown population mean μ and variance σ^2 . The null and alternative hypotheses are $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$. To test H_0 , we calculate the sample mean $\bar{X}$ and its standard error, and compute the test statistic $t = \frac{\bar{X} - \mu_0}{\text{s.e.}}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

This sequence describes the testing of a hypotheses relating to regression coefficients. We start by summarizing the basic procedure for a t test discussed in the Review chapter.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

REVIEW CHAPTER

A random variable X is distributed with unknown population mean μ and variance σ^2 . The null and alternative hypotheses are $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$. To test H_0 , we calculate the sample mean $\bar{X}$ and its standard error, and compute the test statistic $t = \frac{\bar{X} - \mu_0}{\text{s.e.}}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

For the theory behind the t test, see the Review chapter. We will summarize only the procedure here.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

REVIEW CHAPTER

A random variable X is distributed with unknown population mean μ and variance σ^2 . The null and alternative hypotheses are $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$. To test H_0 , we calculate the sample mean $\bar{X}$ and its standard error, and compute the test statistic $t = \frac{\bar{X} - \mu_0}{\text{s.e.}}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

We have a random variable X with unknown population mean μ and variance σ^2 . We wish to test a null hypothesis $H_0: \mu = \mu_0$ against the alternative $H_1: \mu \neq \mu_0$, given a sample of n observations.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

REVIEW CHAPTER

A random variable X is distributed with unknown population mean μ and variance σ^2 . The null and alternative hypotheses are $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$. To test H_0 , we calculate the sample mean $\bar{X}$ and its standard error, and compute the test statistic $t = \frac{\bar{X} - \mu_0}{\text{s.e.}}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

We calculate the sample mean $\bar{X}$ and its standard error, and compute the t statistic shown. We reject H_0 if the absolute value of the t statistic is greater than the critical value of t , given the chosen significance level for the test.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

REVIEW CHAPTER

A random variable X is distributed with unknown population mean μ and variance σ^2 . The null and alternative hypotheses are $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$. To test H_0 , we calculate the sample mean $\bar{X}$ and its standard error, and compute the test statistic $t = \frac{\bar{X} - \mu_0}{\text{s.e.}}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

REGRESSION MODEL

Assuming that the true relationship is $Y = \beta_1 + \beta_2 X + u$ and that, given a sample of data, we have fitted the model $\hat{Y} = b_1 + b_2 X$. The null and alternative hypotheses are

$$H_0: \beta_2 = \beta_2^0, H_1: \beta_2 \neq \beta_2^0$$

To test H_0 , we compute $t = \frac{b_2 - \beta_2^0}{\text{s.e.}(b_2)}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

In the case of the regression model, we are concerned with testing hypotheses relating to the unknown parameters of the relationship. Here also we perform t tests. The theory in principle is exactly the same and the procedure is parallel in an obvious way.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

REVIEW CHAPTER

A random variable X is distributed with unknown population mean μ and variance σ^2 . The null and alternative hypotheses are $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$. To test H_0 , we calculate the sample mean $\bar{X}$ and its standard error, and compute the test statistic $t = \frac{\bar{X} - \mu_0}{\text{s.e.}}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

REGRESSION MODEL

Assuming that the true relationship is $Y = \beta_1 + \beta_2 X + u$ and that, given a sample of data, we have fitted the model $\hat{Y} = b_1 + b_2 X$. The null and alternative hypotheses are

$$H_0: \beta_2 = \beta_2^0, H_1: \beta_2 \neq \beta_2^0$$

To test H_0 , we compute $t = \frac{b_2 - \beta_2^0}{\text{s.e.}(b_2)}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

The t statistic is the difference between the estimated coefficient and its hypothesized value, divided by the standard error of the coefficient. We reject the null hypothesis if the absolute value is greater than the critical value of t , given the chosen significance level.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

REVIEW CHAPTER

A random variable X is distributed with unknown population mean μ and variance σ^2 . The null and alternative hypotheses are $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$. To test H_0 , we calculate the sample mean $\bar{X}$ and its standard error, and compute the test statistic $t = \frac{\bar{X} - \mu_0}{\text{s.e.}}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

REGRESSION MODEL

Assuming that the true relationship is $Y = \beta_1 + \beta_2 X + u$ and that, given a sample of data, we have fitted the model $\hat{Y} = b_1 + b_2 X$. The null and alternative hypotheses are

$$H_0: \beta_2 = \beta_2^0, H_1: \beta_2 \neq \beta_2^0$$

To test H_0 , we compute $t = \frac{b_2 - \beta_2^0}{\text{s.e.}(b_2)}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

There is one important difference. When locating the critical value of t , one must take account of the number of degrees of freedom. In the case of the random variable X , this is $n - 1$, where n is the number of observations in the sample.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

REVIEW CHAPTER

A random variable X is distributed with unknown population mean μ and variance σ^2 . The null and alternative hypotheses are $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$. To test H_0 , we calculate the sample mean $\bar{X}$ and its standard error, and compute the test statistic $t = \frac{\bar{X} - \mu_0}{\text{s.e.}}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

REGRESSION MODEL

Assuming that the true relationship is $Y = \beta_1 + \beta_2 X + u$ and that, given a sample of data, we have fitted the model $\hat{Y} = b_1 + b_2 X$. The null and alternative hypotheses are

$$H_0: \beta_2 = \beta_2^0, H_1: \beta_2 \neq \beta_2^0$$

To test H_0 , we compute $t = \frac{b_2 - \beta_2^0}{\text{s.e.}(b_2)}$. We reject H_0 if $|t| > t_{\text{crit}}$ for a chosen significance level.

In the case of the regression model, the number of degrees of freedom is $n - k$, where n is the number of observations in the sample and k is the number of parameters (β coefficients). For the simple regression model above, it is $n - 2$.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

Example:

$$p = \beta_1 + \beta_2 w + u$$

Null hypothesis:

$$H_0: \beta_2 = 1.0$$

Alternative hypothesis:

$$H_1: \beta_2 \neq 1.0$$

As an illustration, we will consider a model relating price inflation to wage inflation. p is the percentage annual rate of growth of prices and w is the percentage annual rate of growth of wages.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

Example:

$$p = \beta_1 + \beta_2 w + u$$

Null hypothesis:

$$H_0: \beta_2 = 1.0$$

Alternative hypothesis:

$$H_1: \beta_2 \neq 1.0$$

We will test the hypothesis that the rate of price inflation is equal to the rate of wage inflation. The null hypothesis is therefore $H_0: \beta_2 = 1.0$. (We should also test $\beta_1 = 0$.)

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

Example:

$$p = \beta_1 + \beta_2 w + u$$

Null hypothesis:

$$H_0: \beta_2 = 1.0$$

Alternative hypothesis:

$$H_1: \beta_2 \neq 1.0$$

$$\hat{p} = 1.21 + 0.82w$$

(0.05) (0.10)

Suppose that the regression result is as shown (standard errors in parentheses). Our actual estimate of the slope coefficient is only 0.82. We will check whether we should reject the null hypothesis.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

Example:

$$p = \beta_1 + \beta_2 w + u$$

Null hypothesis:

$$H_0: \beta_2 = 1.0$$

Alternative hypothesis:

$$H_1: \beta_2 \neq 1.0$$

$$\hat{p} = 1.21 + 0.82w$$

(0.05) (0.10)

$$t = \frac{b_2 - \beta_2^0}{\text{s.e.}(b_2)} = \frac{0.82 - 1.00}{0.10} = -1.80.$$

We compute the t statistic by subtracting the hypothetical true value from the sample estimate and dividing by the standard error. It comes to -1.80 .

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

Example:

$$p = \beta_1 + \beta_2 w + u$$

Null hypothesis:

$$H_0: \beta_2 = 1.0$$

Alternative hypothesis:

$$H_1: \beta_2 \neq 1.0$$

$$\hat{p} = 1.21 + 0.82w$$

(0.05) (0.10)

$$t = \frac{b_2 - \beta_2^0}{\text{s.e.}(b_2)} = \frac{0.82 - 1.00}{0.10} = -1.80.$$

$$n = 20; \text{ degrees of freedom} = 18$$

There are 20 observations in the sample. We have estimated 2 parameters, so there are 18 degrees of freedom.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

Example:

$$p = \beta_1 + \beta_2 w + u$$

Null hypothesis:

$$H_0: \beta_2 = 1.0$$

Alternative hypothesis:

$$H_1: \beta_2 \neq 1.0$$

$$\hat{p} = 1.21 + 0.82w$$

(0.05) (0.10)

$$t = \frac{b_2 - \beta_2^0}{\text{s.e.}(b_2)} = \frac{0.82 - 1.00}{0.10} = -1.80.$$

$$n = 20; \text{ degrees of freedom} = 18$$

$$t_{\text{crit},5\%} = 2.101$$

The critical value of t with 18 degrees of freedom is 2.101 at the 5% level. The absolute value of the t statistic is less than this, so we do not reject the null hypothesis.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

$$Y = \beta_1 + \beta_2 X + u$$

In practice it is unusual to have a feeling for the actual value of the coefficients. Very often the objective of the analysis is to demonstrate that Y is influenced by X , without having any specific prior notion of the actual coefficients of the relationship.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

$$Y = \beta_1 + \beta_2 X + u$$

$$H_0 : \beta_2 = 0; \quad H_1 : \beta_2 \neq 0$$

In this case it is usual to define $\beta_2 = 0$ as the null hypothesis. In words, the null hypothesis is that X does not influence Y . We then try to demonstrate that the null hypothesis is false.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

$$Y = \beta_1 + \beta_2 X + u$$

$$H_0 : \beta_2 = 0; \quad H_1 : \beta_2 \neq 0$$

$$t = \frac{b_2 - \beta_2^0}{\text{s.e.}(b_2)} = \frac{b_2}{\text{s.e.}(b_2)}$$

For the null hypothesis $\beta_2 = 0$, the t statistic reduces to the estimate of the coefficient divided by its standard error.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

$$Y = \beta_1 + \beta_2 X + u$$

$$H_0 : \beta_2 = 0; \quad H_1 : \beta_2 \neq 0$$

$$t = \frac{b_2 - \beta_2^0}{\text{s.e.}(b_2)} = \frac{b_2}{\text{s.e.}(b_2)}$$

This ratio is commonly called the t statistic for the coefficient and it is automatically printed out as part of the regression results. To perform the test for a given significance level, we compare the t statistic directly with the critical value of t for that significance level.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540			
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15	
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000	
Total	112010.231	539	207.811189	R-squared	=	0.1725	
				Adj R-squared	=	0.1710	
				Root MSE	=	13.126	

EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

Here is the output from the earnings function fitted in a previous slideshow, with the t statistics highlighted.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540			
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15	
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000	
				R-squared	=	0.1725	
				Adj R-squared	=	0.1710	
Total	112010.231	539	207.811189	Root MSE	=	13.126	

EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

You can see that the t statistic for the coefficient of S is enormous. We would reject the null hypothesis that schooling does not affect earnings at the 1% significance level (critical value about 2.59).

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540			
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15	
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000	
				R-squared	=	0.1725	
				Adj R-squared	=	0.1710	
Total	112010.231	539	207.811189	Root MSE	=	13.126	

EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

In this case we could go further and reject the null hypothesis that schooling does not affect earnings at the 0.1% significance level.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540		
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000
Total	112010.231	539	207.811189	R-squared	=	0.1725
				Adj R-squared	=	0.1710
				Root MSE	=	13.126

EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

The advantage of reporting rejection at the 0.1% level, instead of the 1% level, is that the risk of mistakenly rejecting the null hypothesis of no effect is now only 0.1% instead of 1%. The result is therefore even more convincing.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540		
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000
Total	112010.231	539	207.811189	R-squared	=	0.1725
				Adj R-squared	=	0.1710
				Root MSE	=	13.126

EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

The t statistic for the intercept is also enormous. However, since the intercept does not have any meaning, it does not make sense to perform a t test on it.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540		
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Residual	92688.6722	538	172.283777	Prob > F	=	0.0000
Total	112010.231	539	207.811189	R-squared	=	0.1725
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EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
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_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

The next column in the output gives what are known as the p values for each coefficient. This is the probability of obtaining the corresponding t statistic as a matter of chance, if the null hypothesis $H_0: \beta = 0$ is true.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

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EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

If you reject the null hypothesis $H_0: \beta = 0$, this is the probability that you are making a mistake and making a Type I error. It therefore gives the significance level at which the null hypothesis would just be rejected.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

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Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
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EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

If $p = 0.05$, the null hypothesis could just be rejected at the 5% level. If it were 0.01, it could just be rejected at the 1% level. If it were 0.001, it could just be rejected at the 0.1% level. This is assuming that you are using two-sided tests.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540		
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000
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EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

In the present case $p = 0$ to three decimal places for the coefficient of S . This means that we can reject the null hypothesis $H_0: \beta_2 = 0$ at the 0.1% level, without having to refer to the table of critical values of t . (Testing the intercept does not make sense in this regression.)

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540		
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
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EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

It is a more informative approach to reporting the results of test and widely used in the medical literature.

TESTING A HYPOTHESIS RELATING TO A REGRESSION COEFFICIENT

Two-sided t tests

```
. reg EARNINGS S
```

Source	SS	df	MS	Number of obs = 540		
Model	19321.5589	1	19321.5589	F(1, 538)	=	112.15
Residual	92688.6722	538	172.283777	Prob > F	=	0.0000
Total	112010.231	539	207.811189	R-squared	=	0.1725
				Adj R-squared	=	0.1710
				Root MSE	=	13.126

EARNINGS	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
S	2.455321	.2318512	10.59	0.000	1.999876	2.910765
_cons	-13.93347	3.219851	-4.33	0.000	-20.25849	-7.608444

However in economics standard practice is to report results referring to 5% and 1% significance levels, and sometimes to the 0.1% level (when one can reject at that level).